

Challenge (Habanero)

- Q1.** A hiker's backpack weighs $4x + 5$ pounds. If the backpack weighs 21 pounds, what is x ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** A car travels 250 miles on 10 gallons of gas. If x gallons are used, how many miles are traveled?
- (A) $25x$ miles
(B) $10x$ miles
(C) $20x$ miles
(D) $15x$ miles
(E) $5x$ miles
- Q3.** A park ranger observes that a population of rabbits grows according to the equation $y = 2x + 5$. If $y = 11$, what is x ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** Solve for x : $x + 2 = 9$
- (A) $x = 7$
(B) $x = 11$
(C) $x = 5$
(D) $x = 3$
(E) $x = 1$
- Q5.** A group of friends want to go on a 3-day hike. Their food supply is given by $y = 3x - 2$. If they have 7 days of food, how many days will they actually hike?
- (A) 3 days
(B) 5 days
(C) 7 days
(D) 9 days
(E) 11 days
- Q6.** A car's fuel consumption is given by $y = 2x + 1$. If $y = 5$, what is x ?
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q7.** Solve for x : $4x - 2 = 2x + 6$
- (A) $x = 2$
(B) $x = 4$
(C) $x = 6$
(D) $x = 8$
(E) $x = 10$
- Q8.** A trail's length is given by $y = x + 3$. If the trail is 9 miles long, what is x ?
- (A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Open Ended Questions (FRQ)

- Q9.** A group of hikers are planning a trip. Their bus will travel x miles to the starting point, and then they will hike $2x + 5$ miles to the campsite. If the total distance is 35 miles, find x and show your work.
- Q10.** A car is traveling from City A to City B. The distance between the cities is 240 miles. The car's speed is given by $v = 2x + 10$, where v is in miles per hour and x is in hours. If the trip takes 4 hours, find x and the car's average speed.

Chef's Tips

- Make sure to read each question carefully.
- Check your work for each problem.
- Use a pencil to write your answers.